

## 1. Preprocessing

### ✓ Which preprocessing step would you skip here, and why ?

We would skip stemming, because it often produces distorted tokens (e.g., “suggestions” → “suggest”), and in this case it even caused recursion errors. Lemmatization, on the other hand, gives valid dictionary forms (e.g., “suggestions” → “suggestion”) and is more linguistically accurate. Therefore, skipping stemming avoids errors and preserves better semantic quality for the downstream model.

### ✓ Which stopword would you keep in this task, and why?

We would keep the stopword “not”, because it carries strong semantic toxicity detection. Removing it could invert the polarity of a sentence (e.g., “not good” vs. “good”), which would mislead the model. Furthermore, in order to detect toxicity in comments, it is essential to retain certain stopwords such as “you, your, yours” as they often indicate an attack directed at a specific person.

### ✓ One preprocessing tweak that might help toxic detection and that we have not covered

Before converting the texts to lowercase, we extracted the percentage of capital letters, exclamation marks, and question marks—already captured in the created variables—to capture these signals of intensity and emotion in additional columns, enriching the final text vector (e.g., “YOU ARE STUPID”). This information saved, we applied the conversion to lowercase and removed punctuation, reducing the number of features while retaining as much relevant information as possible. Normalizing obfuscated words – users frequently replace letters with symbols (e.g., “f\*ck”) – by mapping such characters back to their alphabet equivalents ensures that toxic words are not missed by the model.

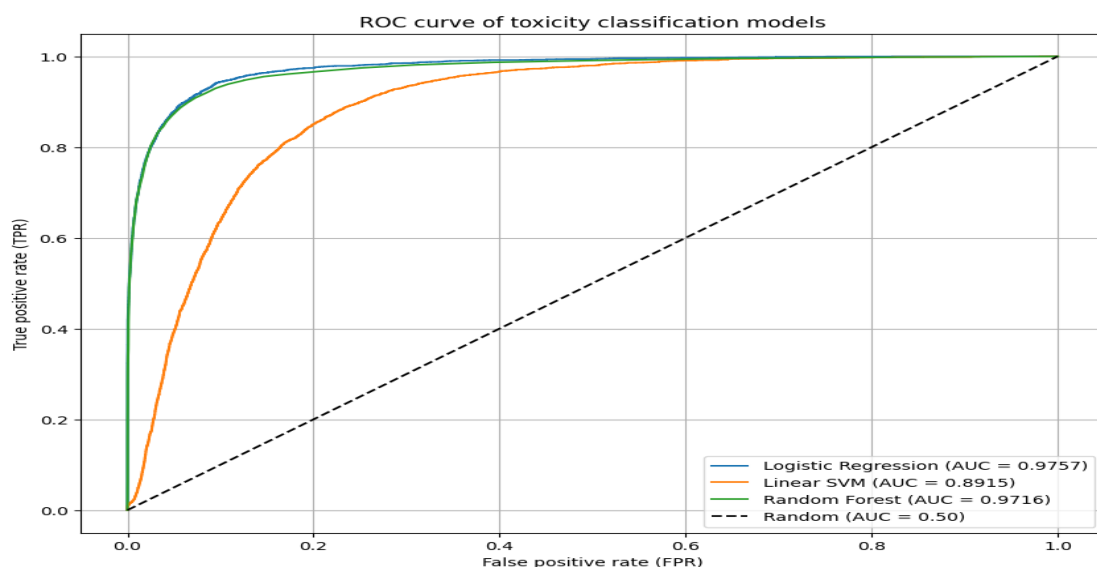
## 2. Feature Engineering

For toxic language detection, bigrams (2-grams) are often more useful than unigrams. While unigrams capture single toxic words, bigrams can detect short phrases that express toxicity, including negations or multi-word insults like “not good”, “shut up”, or “f\*\*\* off”. These combinations carry more context and reduce misclassification that could occur if words are interpreted individually.

## 3. Modeling and Evaluation

| Model              | Precision | Recall | F1-Score | AUC    |
|--------------------|-----------|--------|----------|--------|
| LogisticRegression | 0.96      | 0.96   | 0.96     | 0.9757 |
| Linear SVM         | 0.91      | 0.84   | 0.86     | 0.8915 |
| Random Forest      | 0.95      | 0.95   | 0.94     | 0.9716 |
| Baseline           | 0.89      | 0.54   | 0.62     |        |

### ROC Curve for models



#### 4. Reflection

##### ✓ Which metric is most important ?

In this context of toxicity detection, **precision** and **recall** are the most important metrics, as overall accuracy is misleading due to class imbalance. Recall is the most essential because it indicates the proportion of actually toxic comments correctly identified by the model.

##### ✓ Consequences of each type of error

The two types of errors have different consequences and affect different people.

**False Positive (FP)** : The commenter sees their comment deleted or is warned by mistake, which can lead to frustration or a feeling of censorship. Moderators waste time reviewing a comment that was not a problem and **People affected** are the commenter, moderators, and the platform.

**False Negative (FN)** : The person targeted by the comment and the community are exposed to harmful content. The platform's environment becomes hostile, which can drive users away and **People affected** are Users (the victim and other readers), moderators (who have to deal with the problem later), and the platform (which loses quality and reputation).

##### ✓ Impact of the model's usage context

The choice of the most important metric depends on the model's application.

**For an automatic blocking system** : **Precision** is most important. A false positive (wrongly blocking a user or comment) would have very serious consequences for the platform's reputation and user trust. It is better to let a few toxic comments pass than to wrongly censor a user.

**For a moderation support tool** : **Recall** is more important. The goal is to help moderators quickly spot as many problematic comments as possible. A false positive is only a minor inconvenience (a moderator must check a non-toxic comment), whereas a false negative (a missed toxic comment) would be a missed opportunity to protect the community.

##### ✓ Analogy with COVID testing

The analogy with COVID testing illustrates the trade-off between errors well:

**False Positive** : A healthy person tests positive.

**Consequences** : The person quarantines unnecessarily, experiences stress, and may miss work. The impact is primarily individual.

**False Negative** : A sick person tests negative.

**Consequences** : The person thinks they are healthy and continues to interact with others, spreading the virus. The impact is on the public health of the community.

**Similarities to Wikipedia moderation**: In both cases, the **false negative is often more dangerous for the community**. Letting a toxic comment or a sick person interact freely can cause significant damage to the environment as a whole, although false positives are also important. The severity of the false negative's consequence is what makes **recall** so critical in both contexts.